

Social Media Analysis

(Instagram Version)

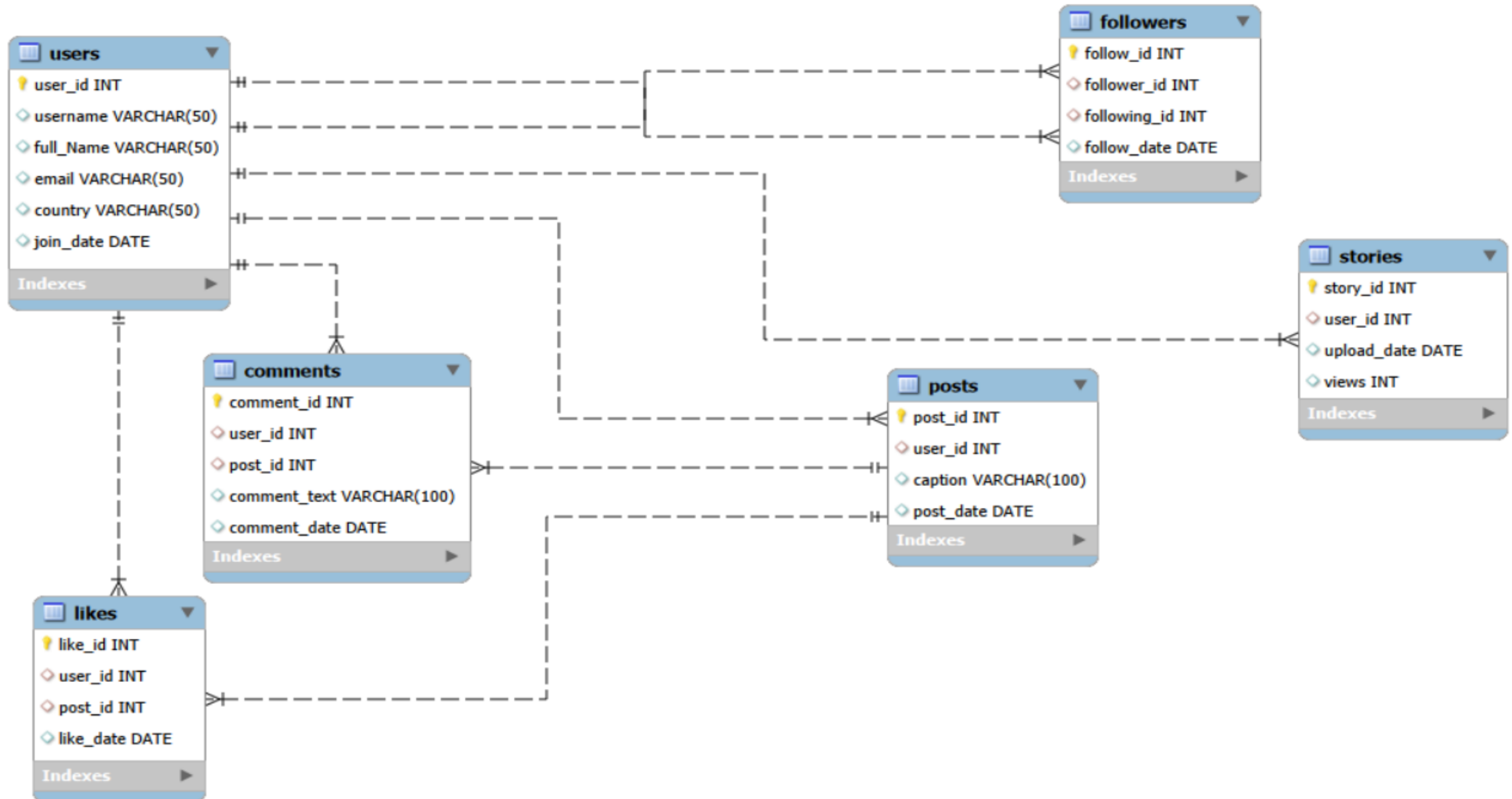


Abstract

- *This Project “Social Media Analysis (Instagram Version)” Is Made To Understand A Real-world Social Media Platform Using Mysql.*
- *This Project Focus To Analyze Social Media Data (Instagram).*
- *The Database Includes Tables Like : Users, Posts, Likes, Comments, Followers, Stories. Also It Helps To Understand User Behavior, Engagement, Patterns And Trending Content.*



ER-DIAGRAM



STRUCTURE OF TABLE

❖ USER'S TABLE

*This Table (Users) Shows
User's ID , User Names ,
User (His/Her) Full Name , Email ID ,
And Which Country They Are Living
Also Shows Join Date when They Join
Social Media Platform .*

Syntax : Desc Users ;

```
mysql> desc users;
```

Field	Type	Null	Key	Default	Extra
user_id	int	NO	PRI	NULL	
username	varchar(50)	YES		NULL	
full_Name	varchar(50)	YES		NULL	
email	varchar(50)	YES		NULL	
country	varchar(50)	YES		NULL	
join_date	date	YES		NULL	

```
6 rows in set (0.00 sec)
```

STRUCTURE OF TABLE

❖ POST'S TABLE

*This Table (Posts) Shows
Post ID , User ID ,
Caption's which They Are Write
While Posting ,
Also Shows The Post Date On
Which Date They Post On
Instagram .*

Syntax : Desc Posts ;

```
mysql> desc posts ;
```

Field	Type	Null	Key	Default	Extra
post_id	int	NO	PRI	NULL	
user_id	int	YES	MUL	NULL	
caption	varchar(100)	YES		NULL	
post_date	date	YES		NULL	

```
4 rows in set (0.00 sec)
```

STRUCTURE OF TABLE

❖ *LIKE'S TABLE*

*This Table (Likes) Shows
Likes ID, User ID ,
Post ID ,
Also Shows The Like Date On
Which Date They Like Post On
Instagram .*

Syntax : Desc Likes ;

```
mysql> Desc Likes ;
```

Field	Type	Null	Key	Default	Extra
like_id	int	NO	PRI	NULL	
user_id	int	YES	MUL	NULL	
post_id	int	YES	MUL	NULL	
like_date	date	YES		NULL	

```
4 rows in set (0.00 sec)
```

STRUCTURE OF TABLE

❖ COMMENT'S TABLE

*This Table (Comments) Shows
Comments ID , User ID ,
Post ID , And Shows The Comments ,
Also Display The Comment Date .*

Syntax : Desc Comments ;

```
mysql> Desc Comments ;
```

Field	Type	Null	Key	Default	Extra
comment_id	int	NO	PRI	NULL	
user_id	int	YES	MUL	NULL	
post_id	int	YES	MUL	NULL	
comment_text	varchar(100)	YES		NULL	
comment_date	date	YES		NULL	

```
5 rows in set (0.00 sec)
```


STRUCTURE OF TABLE

❖ FOLLOWER'S TABLE

*This Table (Followers) Shows
Follow ID , Follower ID (Which
User Is Following) , Following ID ,
And Show Follow Date On Which
Date The Follow .*

Syntax : Desc Followers ;

```
mysql> Desc Followers ;
+-----+-----+-----+-----+-----+
| Field          | Type  | Null  | Key  | Default | Extra |
+-----+-----+-----+-----+-----+
| follow_id      | int   | NO    | PRI  | NULL    |       |
| follower_id    | int   | YES   | MUL  | NULL    |       |
| following_id    | int   | YES   | MUL  | NULL    |       |
| follow_date    | date  | YES   |      | NULL    |       |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```


STRUCTURE OF TABLE

❖ *STORIE'S TABLE*

*This Table (Stories) Shows
Story ID , User ID,
Upload Date On Which Date User
Upload Story And How Many View
He Get .*

Syntax : Desc Stories ;

```
mysql> Desc Stories ;
```

Field	Type	Null	Key	Default	Extra
story_id	int	NO	PRI	NULL	
user_id	int	YES	MUL	NULL	
upload_date	date	YES		NULL	
views	int	YES		NULL	

```
4 rows in set (0.00 sec)
```

CONTENT
OF
USER'S
TABLE

user_id	username	full_Name	email	country	join_date
1	user1	Amit Sharma	amit1@gmail.com	India	2023-01-01
2	user2	Neha Verma	neha2@gmail.com	India	2023-01-02
3	user3	Raj Patel	raj3@gmail.com	India	2023-01-03
4	user4	Pooja Singh	pooja4@gmail.com	India	2023-01-04
5	user5	Karan Mehta	karan5@gmail.com	India	2023-01-05
6	user6	Anjali Das	anjali6@gmail.com	India	2023-01-06
7	user7	Vikas Gupta	vikas7@gmail.com	India	2023-01-07
8	user8	Sneha Iyer	sneha8@gmail.com	India	2023-01-08
9	user9	Rohit Sharma	rohit9@gmail.com	India	2023-01-09
10	user10	Priya Nair	priya10@gmail.com	India	2023-01-10
11	user11	Arjun Reddy	arjun11@gmail.com	India	2023-01-11
12	user12	Meera Joshi	meera12@gmail.com	India	2023-01-12
13	user13	Rahul Roy	rahul13@gmail.com	India	2023-01-13
14	user14	Divya Kapoor	divya14@gmail.com	India	2023-01-14
15	user15	Suresh Yadav	suresh15@gmail.com	India	2023-01-15
16	user16	Kavita Jain	kavita16@gmail.com	India	2023-01-16
17	user17	Aman Khan	aman17@gmail.com	India	2023-01-17
18	user18	Ritu Malhotra	ritu18@gmail.com	India	2023-01-18
19	user19	Deepak Sharma	deepak19@gmail.com	India	2023-01-19
20	user20	Nikita Singh	nikita20@gmail.com	India	2023-01-20
21	user21	Varun Dhawan	varun21@gmail.com	India	2023-01-21
22	user22	Alok Nath	alok22@gmail.com	India	2023-01-22
23	user23	Isha Gupta	isha23@gmail.com	India	2023-01-23
24	user24	Manoj Bajpayee	manoj24@gmail.com	India	2023-01-24
25	user25	Sonam Kapoor	sonam25@gmail.com	India	2023-01-25
26	user26	Rakesh Roshan	rakesh26@gmail.com	India	2023-01-26
27	user27	Salman Sheikh	salman27@gmail.com	India	2023-01-27
28	user28	Farah Khan	farah28@gmail.com	India	2023-01-28
29	user29	Anushka Sharma	anushka29@gmail.com	India	2023-01-29
30	user30	Simran Kaur	simran30@gmail.com	India	2023-01-30

*CONTENT
OF
POST'S
TABLE*

post_id	user_id	caption	post_date
1	1	Sunset view	2023-02-01
2	2	New car	2023-02-02
3	3	Vacation	2023-02-03
4	4	Food time	2023-02-04
5	5	Gym day	2023-02-05
6	6	Nature love	2023-02-06
7	7	Beach vibes	2023-02-07
8	8	Selfie	2023-02-08
9	9	Cricket match	2023-02-09
10	10	Office day	2023-02-10
11	1	Morning walk	2023-02-11
12	2	Shopping	2023-02-12
13	3	Friends	2023-02-13
14	4	Movie night	2023-02-14
15	5	Workout	2023-02-15
16	6	Travel	2023-02-16
17	7	Hiking	2023-02-17
18	8	Party	2023-02-18
19	9	Family time	2023-02-19
20	10	Coding	2023-02-20
21	11	Music	2023-02-21
22	12	Dance	2023-02-22
23	13	Festival	2023-02-23
24	14	Wedding	2023-02-24
25	15	Driving	2023-02-25
26	16	Books	2023-02-26
27	17	Art	2023-02-27
28	18	Shopping	2023-02-28
29	19	Travel	2023-03-01
30	20	Food	2023-03-02

*CONTENT
OF
LIKE'S
TABLE*

like_id	user_id	post_id	like_date
1	2	1	2023-02-03
2	3	1	2023-02-03
3	4	2	2023-02-04
4	5	2	2023-02-04
5	6	3	2023-02-05
6	7	3	2023-02-05
7	8	4	2023-02-06
8	9	4	2023-02-06
9	10	5	2023-02-07
10	1	5	2023-02-07
11	2	6	2023-02-08
12	3	6	2023-02-08
13	4	7	2023-02-09
14	5	7	2023-02-09
15	6	8	2023-02-10
16	7	8	2023-02-10
17	8	9	2023-02-11
18	9	9	2023-02-11
19	10	10	2023-02-12
20	1	10	2023-02-12
21	11	11	2023-02-13
22	12	12	2023-02-14
23	13	13	2023-02-15
24	14	14	2023-02-16
25	15	15	2023-02-17
26	16	16	2023-02-18
27	17	17	2023-02-19
28	18	18	2023-02-20
29	19	19	2023-02-21
30	20	20	2023-02-22

CONTENT
OF
COMMENT'S
TABLE

comment_id	user_id	post_id	comment_text	comment_date
1	3	1	Nice	2023-02-03
2	4	1	Awesome	2023-02-03
3	5	2	Great	2023-02-04
4	6	2	Cool	2023-02-04
5	7	3	Wow	2023-02-05
6	8	3	Super	2023-02-05
7	9	4	Amazing	2023-02-06
8	10	4	Nice shot	2023-02-06
9	1	5	Good	2023-02-07
10	2	5	Love it	2023-02-07
11	3	6	Beautiful	2023-02-08
12	4	6	Nice view	2023-02-08
13	5	7	Cool pic	2023-02-09
14	6	7	Wow	2023-02-09
15	7	8	Awesome	2023-02-10
16	8	8	Nice	2023-02-10
17	9	9	Great	2023-02-11
18	10	9	Cool	2023-02-11
19	1	10	Super	2023-02-12
20	2	10	Nice	2023-02-12
21	11	11	Amazing	2023-02-13
22	12	12	Love	2023-02-14
23	13	13	Nice	2023-02-15
24	14	14	Great	2023-02-16
25	15	15	Cool	2023-02-17
26	16	16	Wow	2023-02-18
27	17	17	Nice	2023-02-19
28	18	18	Awesome	2023-02-20
29	19	19	Great	2023-02-21
30	20	20	Nice	2023-02-22

*CONTENT
OF
FOLLOWERS
TABLE*

follow_id	follower_id	following_id	follow_date
1	2	1	2023-02-01
2	3	1	2023-02-01
3	4	2	2023-02-02
4	5	2	2023-02-02
5	6	3	2023-02-03
6	7	3	2023-02-03
7	8	4	2023-02-04
8	9	4	2023-02-04
9	10	5	2023-02-05
10	1	5	2023-02-05
11	2	6	2023-02-06
12	3	6	2023-02-06
13	4	7	2023-02-07
14	5	7	2023-02-07
15	6	8	2023-02-08
16	7	8	2023-02-08
17	8	9	2023-02-09
18	9	9	2023-02-09
19	10	10	2023-02-10
20	1	10	2023-02-10
21	11	12	2023-02-11
22	12	13	2023-02-12
23	13	14	2023-02-13
24	14	15	2023-02-14
25	15	16	2023-02-15
26	16	17	2023-02-16
27	17	18	2023-02-17
28	18	19	2023-02-18
29	19	20	2023-02-19
30	20	21	2023-02-20

*CONTENT
OF
STORI'S
TABLE*

story_id	user_id	upload_date	views
1	1	2023-02-10	100
2	2	2023-02-11	120
3	3	2023-02-12	130
4	4	2023-02-13	140
5	5	2023-02-14	150
6	6	2023-02-15	160
7	7	2023-02-16	170
8	8	2023-02-17	180
9	9	2023-02-18	190
10	10	2023-02-19	200
11	11	2023-02-20	110
12	12	2023-02-21	120
13	13	2023-02-22	130
14	14	2023-02-23	140
15	15	2023-02-24	150
16	16	2023-02-25	160
17	17	2023-02-26	170
18	18	2023-02-27	180
19	19	2023-02-28	190
20	20	2023-03-01	200
21	21	2023-03-02	210
22	22	2023-03-03	220
23	23	2023-03-04	230
24	24	2023-03-05	240
25	25	2023-03-06	250
26	26	2023-03-07	260
27	27	2023-03-08	270
28	28	2023-03-09	280
29	29	2023-03-10	290
30	30	2023-03-11	300

JOINS



JOINS

*1.Display Each user post Id And
Post Caption
(users + posts+ Caption)*

Syntax :

```
Select  
u.username,  
p.post_id,  
p.Caption  
from users u  
inner join posts p  
on u.user_id =  
p.user_id ;
```

username	post_id	caption
user1	1	Sunset view
user2	2	New car
user3	3	Vacation
user4	4	Food time
user5	5	Gym day
user6	6	Nature love
user7	7	Beach vibes
user8	8	Selfie
user9	9	Cricket match
user10	10	Office day
user1	11	Morning walk
user2	12	Shopping
user3	13	Friends
user4	14	Movie night
user5	15	Workout
user6	16	Travel
user7	17	Hiking
user8	18	Party
user9	19	Family time
user10	20	Coding
user11	21	Music
user12	22	Dance
user13	23	Festival
user14	24	Wedding
user15	25	Driving
user16	26	Books
user17	27	Art
user18	28	Shopping
user19	29	Travel
user20	30	Food

JOINS

2.Display Each User And What They Commented on Other User's Post (users + Comments)

Syntax :

```
Select  
u.username,  
c.comment_text  
from users u  
inner join comments c  
on u.user_id =  
c.user_id ;
```

username	comment_text
user3	Nice
user4	Awesome
user5	Great
user6	Cool
user7	Wow
user8	Super
user9	Amazing
user10	Nice shot
user1	Good
user2	Love it
user3	Beautiful
user4	Nice view
user5	Cool pic
user6	Wow
user7	Awesome
user8	Nice
user9	Great
user10	Cool
user1	Super
user2	Nice
user11	Amazing
user12	Love
user13	Nice
user14	Great
user15	Cool
user16	Wow
user17	Nice
user18	Awesome
user19	Great
user20	Nice

JOINS

3.Display The Follower's As Per Each User (users + followers)

Syntax :

```
select  
  u.username,  
  count( f.follower_id) as  
  followers  
from users u  
left join followers f  
on u.user_id = f.following_id  
group by u.username ;
```

username	followers
user1	2
user2	2
user3	2
user4	2
user5	2
user6	2
user7	2
user8	2
user9	2
user10	2
user11	0
user12	1
user13	1
user14	1
user15	1
user16	1
user17	1
user18	1
user19	1
user20	1
user21	1
user22	0
user23	0
user24	0
user25	0
user26	0
user27	0
user28	0
user29	0
user30	0

JOINS

*4.Display which user received
total likes
(user+ posts + likes)*

Syntax :

```
Select  
u.username,  
count(l.like_id) as  
total_Likes  
from users u  
join posts p on u.user_id =  
p.user_id  
left join likes l on p.post_id  
= l.post_id  
group by username;
```

username	total_Likes
user1	3
user2	3
user3	3
user4	3
user5	3
user6	3
user7	3
user8	3
user9	3
user10	3
user11	0
user12	0
user13	0
user14	0
user15	0
user16	0
user17	0
user18	0
user19	0
user20	0

JOINS

5. Display The Total Num Of Comments On Each User Post (users+ posts + comments)

Syntax :

```
Select  
u.username,  
count(c.comment_id) as  
total_comments  
from users u  
join posts p on u.user_id = p.user_id  
left join comments c on p.post_id =  
c.post_id  
group by u.username;
```

username	total_comments
user1	3
user2	3
user3	3
user4	3
user5	3
user6	3
user7	3
user8	3
user9	3
user10	3
user11	0
user12	0
user13	0
user14	0
user15	0
user16	0
user17	0
user18	0
user19	0
user20	0

SUBQUERYS



SUBQUERYS

1. Display Thos User Who Never Posted On Instagram

Syntax:

```
select username  
from users  
where user_id not in  
(select user_id from posts) ;
```

```
+-----+  
| username |  
+-----+  
| user21   |  
| user22   |  
| user23   |  
| user24   |  
| user25   |  
| user26   |  
| user27   |  
| user28   |  
| user29   |  
| user30   |  
+-----+  
10 rows in set (0.00 sec)
```

SUBQUERYS

2. Display The User Who's Instagram Story Is Most Viewed

Syntax:

```
select user_id , Views  
from stories  
where views =  
(select max(views) from stories) ;
```

+	-----+	-----+
	user_id	views
+	-----+	-----+
	30	300
+	-----+	-----+
1 row in set (0.00 sec)		

SUBQUERYS

3 . Display The Most Like Post On Instagram .

Syntax :

```
select post_id
from likes
group by post_id
order by count(*) desc
limit 5;
```

```
+-----+
| post_id |
+-----+
|         |
|         |
|         |
|         |
|         |
|         |
+-----+
5 rows in set (0.00 sec)
```

SUBQUERYS

4 . Find And Display The Users Who Have No Followers

Syntax:

```
select username  
from users  
where user_id not in  
(select following_id from followers)  
;
```

```
+-----+  
| username |  
+-----+  
| user11  |  
| user22  |  
| user23  |  
| user24  |  
| user25  |  
| user26  |  
| user27  |  
| user28  |  
| user29  |  
| user30  |  
+-----+  
10 rows in set (0.00 sec)
```

SUBQUERYS

5. Find Users With Maximum Posts And Display Them

Syntax:

```
select username  
from users  
where user_id =  
(select user_id  
from posts  
group by user_id  
order by count(*) desc  
limit 1) ;
```

```
+-----+  
| username |  
+-----+  
| user1    |  
+-----+  
1 row in set (0.00 sec)
```



Prepared By

(Sahil Haresh Darole)



Thank You